The documentation and process conversion measures necessary to comply with this revision shall be completed by 29 January 1994

INCH-POUND

MIL-S-19500/391C
29 October 1993
SUPERSEDING
MIL-S-19500/391B
1 August 1983

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES 2N3019, 2N3019S, 2N3057A, AND 2N3700
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, low-power transistors. Three levels of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Type	P_T		V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}
	T_A = +25°C 1/	T_C = +25°C 2/					
	W	W	V dc	V dc	V dc	A dc	°C
2N3019	0.8	5	140	7	80	1	-55 to +175
2N3019S	0.8	5	140	7	80	1	-55 to +175
2N3057A	0.4	1.8	140	7	80	1	-55 to +175
2N3700	0.5	1.8	140	7	80	1	-55 to +175

1/ Derate linearly at 4.6 mW/°C for types 2N3019 and 2N3019S; 2.3 mW/°C for type 2N3057A; and 2.85 mW/°C for type 2N3700 for $T_A \geq$ +25°C.
2/ Derate linearly at 28.6 mW/°C for types 2N3019 and 2N3019S; 10.3 mW/°C for types 2N3057A and 2N3700 for $T_C \geq$ +25°C.

1.4 Primary electrical characteristics.

Limits	h_{FE1}	h_{FE2}	h_{FE3} 1/	$\|h_{fe}\|$	$V_{CE(sat)}$1/
	V_{CE}= 10 V dc I_C = 150 mA dc	V_{CE}= 10 V dc I_C = 0.1 mA dc	V_{CE}= 10 V dc I_C = 10 mA dc	V_{CE}= 10 V dc I_C = 50 mA dc f = 20 MHz	I_C = 150 mA dc I_B = 15 mA dc
					V dc
Min	100	50	90	5	
Max	300	200		20	0.20

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federaland military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 8, T-1A for 2N3019 (TO-5), T-1C for 2N3019S (TO-39), figure 9, T-2A for 2N3700 (TO-46), and T-2B for 2N3057A (TO-18).

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANTX, JANTXV, and JANS only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS	JANTX and JANTXV levels
9	I_{CES1} and h_{FE1}	Not applicable
11	I_{CES1}; h_{FE1}; ΔI_{CES1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = ±15%.	I_{CES1} and h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 & 3 of table I herein; ΔI_{CES1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15%.	Subgroup 2 of table I herein; ΔI_{CES1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15%.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750 (see 4.5).

JANS level - V_{CB} = 10 V dc, P_T = 800 mW (2N3019 and 2N3019S), 400 mW (2N3057A), and 500 mW (2N3700).

JANTX and JANTXV levels - V_{CB} = 60 V dc, P_T = 800 mW (2N3019 and 2N3019S), 400 mW (2N3057A), and 500 mW (2N3700).

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
4	1037	V_{CB} = 10 V dc; 2N3019 and 2N3019S - P_T = 800 mW; 2N3057A - P_T = 400 mW; 2N3700 - P_T = 500 mW at T_A = 30° ±5°C, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles (see 4.5.1). No heat sink or forced-air cooling on the devices shall be permitted.
5	1027	V_{CB} = 10 V dc; T_A = 125°C ±25°C for 96 hours; 2N3019 and 2N3019S - P_T = 800 mW; 2N3057A - P_T = 400 mW; 2N3700 - P_T = 500 mW at T_A = 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°C. Marking legibility requirements shall not apply (see 4.5.1).
5	2037	Test condition A.
6		Not applicable.

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4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
3	1027	V_{CB} = 60 V dc; 2N3019 and 2N3019S - P_T = 800 mW; 2N3057A - P_T = 400 mW; 2N3700 - P_T = 500 mW at T_A = 30° ±5°C. No heat sink or forced-air cooling on the devices shall be permitted.
5		Not applicable.

4.4.4 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end points) shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
2	2036	Test condition E.
6	1026	V_{CB} = 60 V dc; 2N3019 and 2N3019S - P_T = 800 mW; 2N3057A - P_T = 400 mW; 2N3700 - P_T = 500 mW at T_A = 30° ±5°C. No heat sink or forced-air cooling on device shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector-base time constant. This parameter may be determined by applying an rf signal voltage of 1.0 volt (rms) across the collector-base terminals, and measuring the ac voltage drop (V_{eb}) with a high-impedance rf voltmeter across the emitter-base terminals. With f = 79.8 MHz used for the 1.0 volt signal, the following computation applies:

$$r'_b \cdot C_{c(ps)} = 2 \times V_{eb} \quad \text{(millivolts)}$$

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	140		V dc
Emitter to base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc	$V_{(BR)EBO}$	7		V dc
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 30 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		10	nA dc
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE1}	100	300	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE2}	50	200	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE3}	90		
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE4}	50	200	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE5}	15		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$ 1/		0.2	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc pulsed (see 4.5.1)	$V_{CE(sat)2}$ 1/		0.5	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$ 1/		1.1	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES2}		10	µA dc
Low-temperature operation		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE6}	40		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 1 mA dc f = 1 kHz	h_{fe}	80	400	
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 20 MHz	$\lvert h_{fe} \rvert$	5	20	
Input capacitance (output open circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		60	pF
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		12	pF
Noise figure	3246	V_{CE} = 10 V dc; I_C = 100 µA dc; R_g = 1 kΩ; Power bandwidth = 200 Hz	NF		4	dB
Collector to base time constant		V_{CB} = 10 V dc; I_C = 10 mA dc; f = 79.8 MHz (see 4.5.2)	$r'_b C_c$		400	ps
Pulse response		See figure 1	$t_{on} + t_{off}$		30	ns

See footnote at end of table.

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TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = 25°C; t = 10 ms, 1 cycle (see figure 2)				
Test 1 2N3019, 2N3019S 2N3057A, 2N3700		V_{CE} = 10 V dc; I_C = 500 mA dc I_C = 180 mA dc				
Test 2 2N3019, 2N3019S 2N3057A, 2N3700		V_{CE} = 40 V dc; I_C = 125 mA dc I_C = 45 mA dc				
Test 3 2N3019, 2N3019S 2N3057A, 2N3700		V_{CE} = 80 V dc; I_C = 60 mA dc I_C = 22.5 mA dc				
Electrical measurements		See table II, steps 2, 4, and 7.				
Subgroups 6 and 7						
Not applicable						

For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements. 2/, 3/, 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		10	nA dc
2.	Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		20	nA dc
3.	Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		10	nA dc
4.	Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$ 1/		0.20	V dc
5.	Base emitter voltage (saturated)	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$ 1/		1.1	V dc
6.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE2}	50	200	
7.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc	h_{FE1}	100	300	
8.	Collector-emitter cutoff current	3041	Bias condition D; V_{CE} = 90 V dc; pulsed (see 4.5.1)	ΔI_{CES1}	±100% of initial value or 8 nA dc, whichever is greater.		
9.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc; pulsed (see 4.5.1)	Δh_{FE2}	±25% change from initial reading		
10.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	Δh_{FE1}	±25% change from initial reading		
11.	Collector-emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$ 1/	±50% mV dc change from previous measured value.		

See footnotes at end of table.

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TABLE II. Groups B and C electrical measurements - Continued. 2/ 3/ 4/

1/ Measured within 0.125 inch from the body of the device.

2/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, 5, 6, and 7.

b. Subgroup 4, see table II herein, steps 2, 3, 4, 5, 6, 7, and 11.

c. Subgroup 5, see table II herein, steps 2, 3, 4, 5, 7, 8, and 9.

3/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 6.

b. Subgroups 3 and 6, see table II herein, steps 2 and 9.

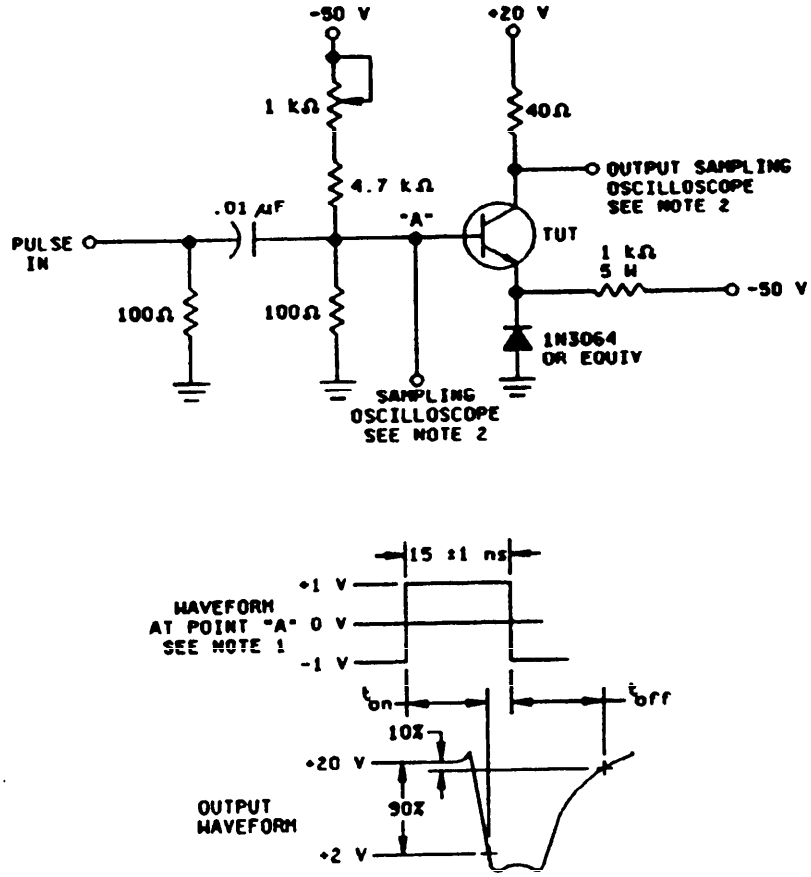
4/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, 4, 5, 6, and 7 (JANS) and steps 1 and 7 (JANTX and JANTXV).

b. Subgroup 3, see table II herein, steps 1, 3, 4, 5, 6, and 7 (JANS) and steps 1 and 7 (JANTX and JANTXV).

c. Subgroup 6, see table II herein, steps 1, 3, 4, 5, 6, 7, 8, and 9 (JANS) and steps 2 and 9 (JANTX and JANTXV).

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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ .2 ns, duty cycle ≤ 2% and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ .02 ns.

FIGURE 1. Nonsaturated switching-time test circuit.

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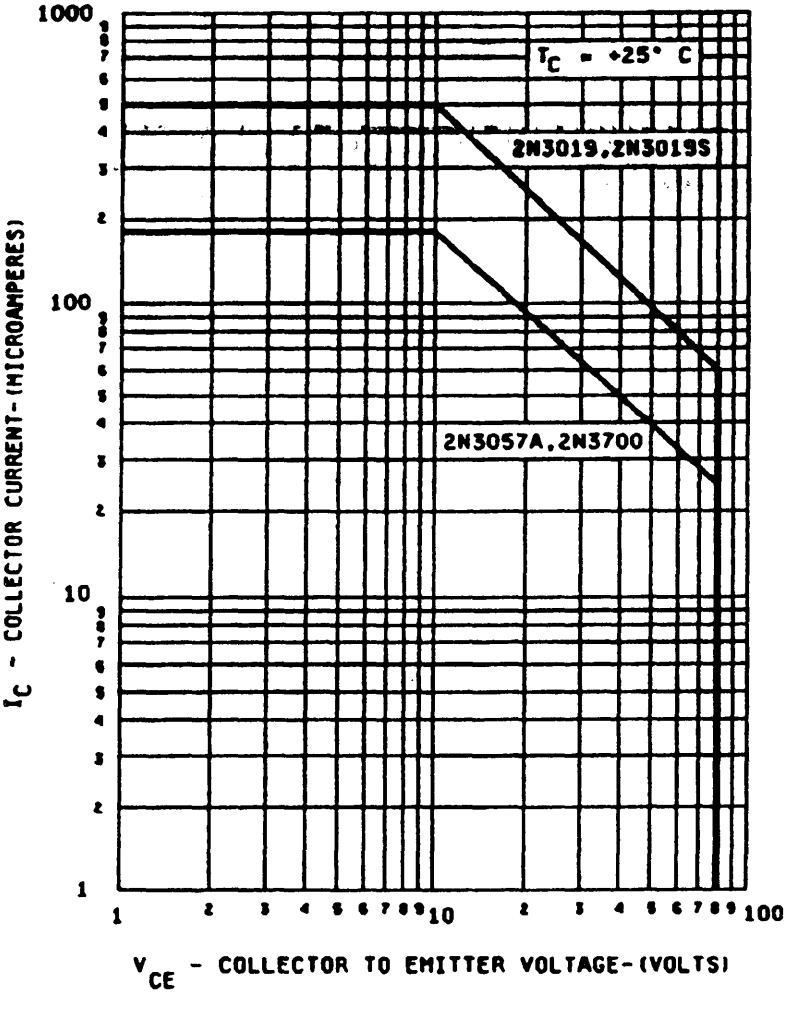


FIGURE 2. Maximum safe operating graph (DC).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish (see 3.3.1)

c. Product assurance level and type designator.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/391C	2. DOCUMENT DATE (YYMMDD) 93/10/29

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER TYPES 2N3019, 2N3019S, 2N3057A, AND 2N3700 JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON 301-731-8603
c. ADDRESS (Include Zip Code) NASA/Parts Projects Office (NPPO) NASA Goddard Space Flight Center Code 310.A Greenbelt, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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